

Answers

1. **C** 2. **B** 3. **B** 4. **B** 5. **D** 6. **D** 7. **A** 8. **A**
9. **D** 10. **C** 11. **D** 12. **C** 13. **A** 14. **C**

Explanations

1. **C**

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number, the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians = 24

Erdos number at the third day: $f+1, b, f+1, d, e, f, g, h$

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians = 20 So here the difference of 4 [24-20] arose, which means e will be $f+5$ initially and changed to $f+1$ after co-authoring with F.

So the Erdos number at the third day: $f+1, b, f+1, d, f+5, f, g, h$

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be $f+5$ because then there will be two mathematicians with the same Erdos number $f+1$.

So five mathematicians will have $f+1$, one with $f+5$, one with f , one with some different value say x

$$5(f+1) + f + 5 + f + x = 24$$

$$7f + x = 14$$

The only value which satisfies the above equation is $f=1, x=7$

Erdos number at the end of fifth day, $f+1, b, f+1, d, f+1, f, g, h$ On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	$f+1$	$f+1$
B	b	b	b
C	c	$f+1$	$f+1$
D	d	d	d
E	e	$e = f+5$	$f+1$
F	$f(\text{min})$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Hence erdos no. of E at the beginning of conference would be $f+5 = 6$.

 VIDEO SOLUTION

2. **B**

4. B

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number ,the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos value then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3

Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day: $f+1,b,f+1,d,e,f,g,h$

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So here the difference of $4[24-20]$ arose, which means e will be $f+5$ initially and changed to $f+1$ after co-authoring with F.

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$$7f+x=14$$

The only value which satisfies the above equation is $f=1,x=7$

Erdos number at the end of fifth day, $f+1,b,f+1,d,f+1,f,g,h$

On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
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E	e	$e = f+5$	$f+1$
F	$f(\text{min})$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Hence the person having the largest Erdos number at the end of the conference must have had Erdos number 7 . Hence option B.

5. D

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number ,the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day: $f+1, b, f+1, d, e, f, g, h$

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$$5(f+1) + f + 5 + f + x = 24$$

$$7f + x = 14$$

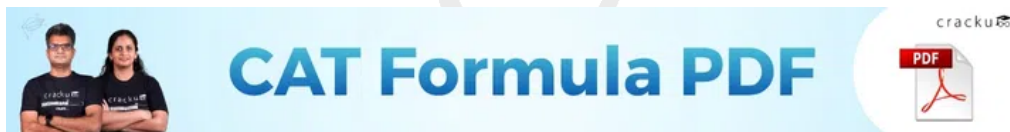
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Erdos number at the end of fifth day, $f+1, b, f+1, d, f+1, f, g, h$ On tabulating, we get

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F	$f(\min)$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

So B,D ,F,G,H are 5 participants in the conference who did not change their Erdos number during the conference.

VIDEO SOLUTION



6. D

Considering option,

In 1st we get product of a and b which is not required In 2nd we get fraction 1.5 which doesn't help in finding average In 3rd we get $(a+b)/4$ after 2 operations which doesn't give average in further operations. only in 4th option we get overall and exact average of A, B and C .

$$/(X(@ (/ (@ (B, A), 2), C), 3), 2) = /(X(@ (/ (A+B)/2), 2), C), 3), 2)$$

$$= /(X(@ (A+B), C), 3), 2) = /(X((A+B, C)/2, 3), 2)$$

$$= /[(A+B+C)/6, 2]$$

$$= (A+B+C)/3$$

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7. A

Considering 1st option $/(@ (A, B), 2)$ First operation is average of A and B so we get $(a+b)/2$ and then we in next operation we multiply it by 2. So we get addition of A and B . So option A

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8. **A**

If the number of mynahs is 7 or more and if 7 birds escaped, there is a possibility that no pigeon escaped. But the bird-keeper was able to conclude that atleast one pigeon escaped. So, the number of mynahs is less than 7.

From the options, we can see that only option a) violates this condition.

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9. **D**

We dont know out of other 2 doctors which doctor is specialist in which disease.

So it is not possible to find out the exact order.

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10. **C**

All of them can be mislabeled in 2 ways:

Red box - white label

White box - Red and white label

Red and white box - Red label

Red box - red and white label

White box - Red label

Red and white box - White label

So, we would try the box with the red and white label and if it has a white ball, labelling to the boxes is done as per case 1. If it has a red ball labelling is done as per case 2.

Note: It's not a good idea to try the white label as if we get a red ball, we can't make out if we are picking from red box or red and white box. Similarly if we try from the box with red label and we get a white ball, again we can't make out if it is coming from the white box or the red and white box.

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A blue banner for 'IIM Call Predictor'. On the left, there is a photo of two people, a man and a woman, both wearing black t-shirts with 'IIM' on them. To their right, the text 'IIM Call Predictor' is written in large white font. Below this, in a smaller yellow box, it says 'Accurate analysis of your profile'.

11. **D**

First counting only nos. 1,2 and 3.

So, we have 6 mangoes, 6 oranges and 7 apples.

We have 4 times 4 so finally we have 2 mangoes, 2 oranges and 7 apples, which is a total of 11 fruits.

[VIEW SOLUTION](#)

12. **C**

On counting only numbers 1,2 and 3, we have 6 mangoes, 6 oranges and 7 apples.

We have 4 times number 4 => Finally we have 2 mangoes , 2 oranges and 7 apples. So, a total of 11 fruits.

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13. **A**

$$((1\#1)\#2) = 2\#2 = 4$$

$$\log_{10} 0.1 = \log_{10} 10^{-1} = -1$$

As $(10^{1.3} \times -1)$ is not positive, $(10^{1.3} \nabla \log_{10} 0.1) = 1$

$$1 \nabla 2 = (1 \times 2)^{(1+2)} = 2^3 = 8$$

Hence, the fraction is $(4-1)/8 = 3/8$

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14. **C**

$$2\#1 = 2 + 1 = 3$$

$$1 \nabla 2 = (1 \times 2)^{(1+2)} = 2^3$$

So answer will be $\frac{3}{8}$

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